

Auction-Flow Strategy Research

Phase A1 findings: leveraged-ETF rebalancing flow, and a more promising lead in the ETFs' own closing auctions

Bottom line

- **The trade as specified in the handoff does not pass on free data.** Fading single-stock leveraged-ETF rebalancing overnight shows no net reversal by the next close. The price pressure builds from ~1:30 pm and is absorbed *before* the close. What's left overnight is small and concentrated in ~10 theme-wide sell-off days.
- **A promising lead turned up in the leveraged ETFs' own closing prices.** On big days the ETF closes away from its NAV. The part of that gap that forms in the last 30 minutes largely reverses by 10:30 the next morning (25-37 bps on large events, in every listing venue).
- **This is the 2007 trade in a new, small venue.** Thin ETF closing auctions absorb price-insensitive closing flow, while fair value (NAV = leverage × the underlying's closing price) is known exactly at 4:00.
- **The key unknown is executability.** Free data cannot tell a real auction imbalance from closing-price noise (bid-ask bounce in thin ETFs). The next step is cheap: check the bid/ask midpoint at 4:00 via IBKR, then auction prints.
- **Statistical honesty:** ~50 tests run in total. The last six months (from 23 Mar 2026) are held out and untouched.

	Strategy A: fade rebalance overnight	New lead: LETF closing premium
Signal	Rebalance flow / daily volume	ETF close vs NAV (premium change)
Effect found	~0 by next close; +6 ± 6 bps overnight ex top-10 days	25-37 bps reversion by next morning on $ e \geq 50$ bps
Robustness	Concentrated (72-86% from 10 days); decaying	All venues, all fund sizes; top-10 days 22-35% of P&L
Status	Does not meet pre-committed bar	Promising; executability unverified

How to read this report: sections 1-2 cover the data and the test of the original idea; section 3 covers the new lead; section 4 lists what could make it wrong; section 5 proposes next steps. All figures use in-sample data only (July 2022 to 20 March 2026).

1. What was built

Everything uses free data (SEC EDGAR, issuer websites, Yahoo Finance). Each piece was validated before use.

- **Fund universe.** Built from the SEC's yearly fund lists (2022-2026), including funds that later closed: **286 daily-reset single-stock leveraged/inverse ETFs on 160 stocks**. Each fund's leverage was checked by regressing its daily returns on its stock's. This caught reused tickers (e.g. NVDB, TSLI), SEC ticker errors, and leverage changes (TSLQ -1x to -2x, MSTX 1.75x to 2x).
- **Fund assets.** 2,062 SEC N-PORT filings give quarter-end assets and monthly flows. GraniteShares and ProShares give exact daily assets. Against the exact data, the reconstruction's median error is **4.6%**.
- **Prices.** Daily prices for ~5,700 US stocks; hourly bars from Oct 2023 for the ~160 stocks, 600 comparison stocks and ~240 leveraged ETFs. 2.7 million stock-days in total.
- **Guardrails.** The six-month hold-out is enforced in code. Earnings days are excluded. Standard errors are clustered by date (and by stock in regressions).

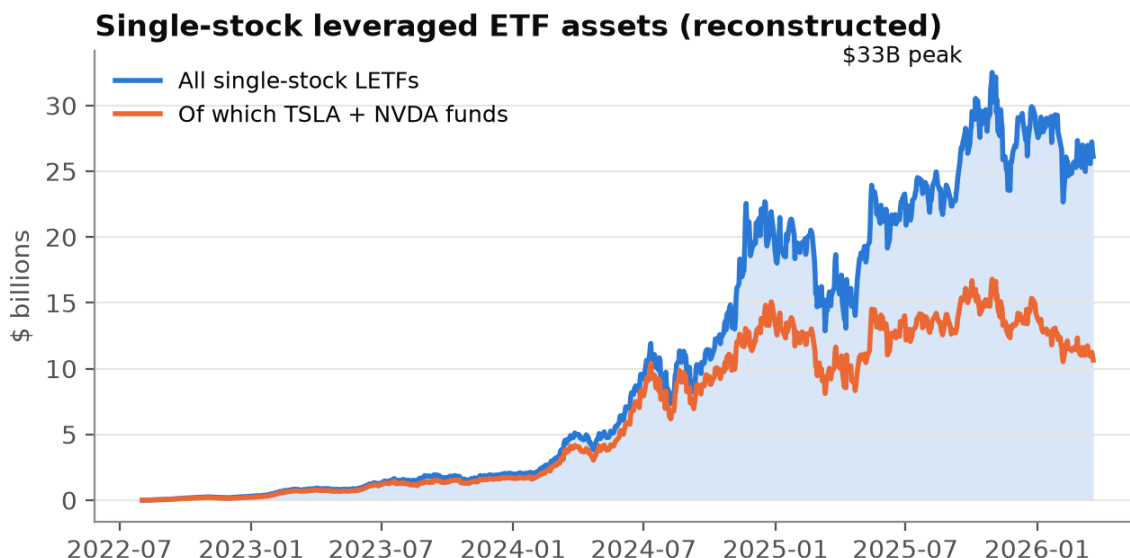


Figure 1. Reconstructed assets of US single-stock leveraged ETFs. Growth came mostly after mid-2024; TSLA and NVDA funds are about half of the total.

2. The original idea: fade the rebalance overnight

Leveraged ETFs must buy after up days and sell after down days, near the close. The hypothesis was that this pushes closing prices away from fair value and reverses overnight. The test sorted stock-days by predicted rebalance flow as a fraction of daily volume, controlling for the stock's own move.

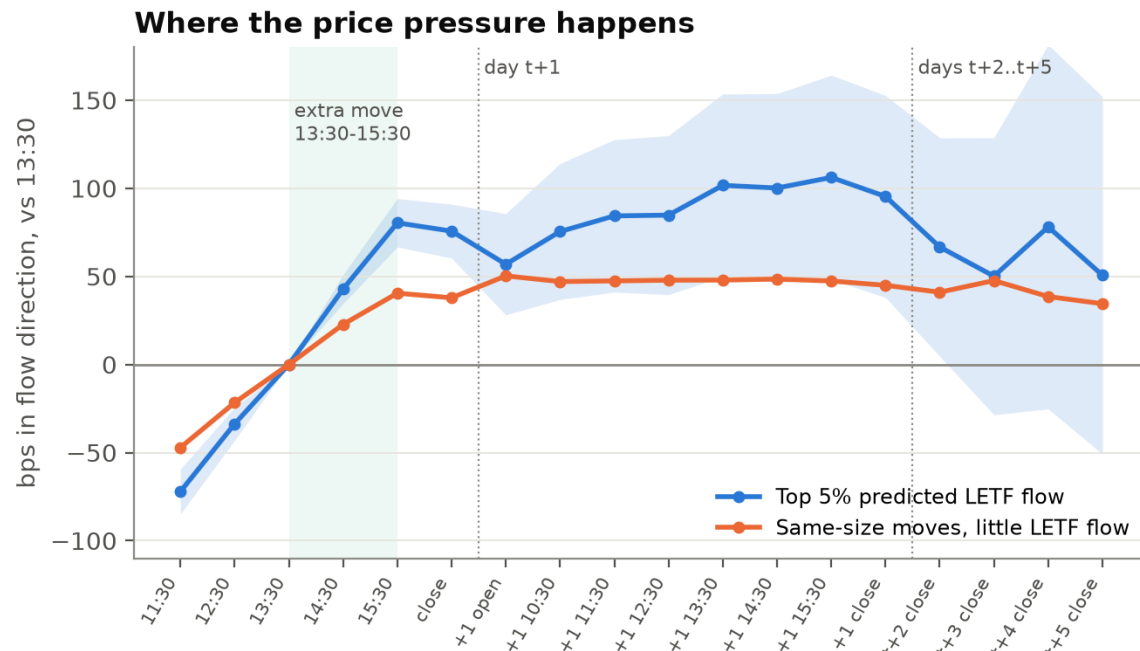


Figure 2. Average price path on the 5% of days with the largest predicted flow (blue) vs. equally large moves in stocks with little leveraged-ETF flow (orange), signed in the flow direction and relative to 13:30. The extra move happens between 13:30 and 15:30. There is nothing extra in the last 30 minutes, and no net reversal by the next close. 1,287 events, 45 stocks, Oct 2023 - Mar 2026.

Key results

Test	Result	Verdict
Pre-registered: flow predicts overnight / next-day reversal	$t = -1.8$ overnight; $t = 0.1$ next close	Fails
Pressure in last 30 min (15:30 to close)	$t = -0.3$	None
Top-decile flow vs same-day matched stocks, overnight	+20 bps; $+6 \pm 6$ excl. top-10 days	Concentrated
Same stocks, stock-specific slopes, overnight	$t = -3.1$ to -3.9 , but ~ 1 -2 bps at typical flow	Tiny
Anticipation: flow at 13:30 predicts 13:30-to-close	$t = 2.7$ pooled; $t = 0.9$ within stock	Not LETF-specific

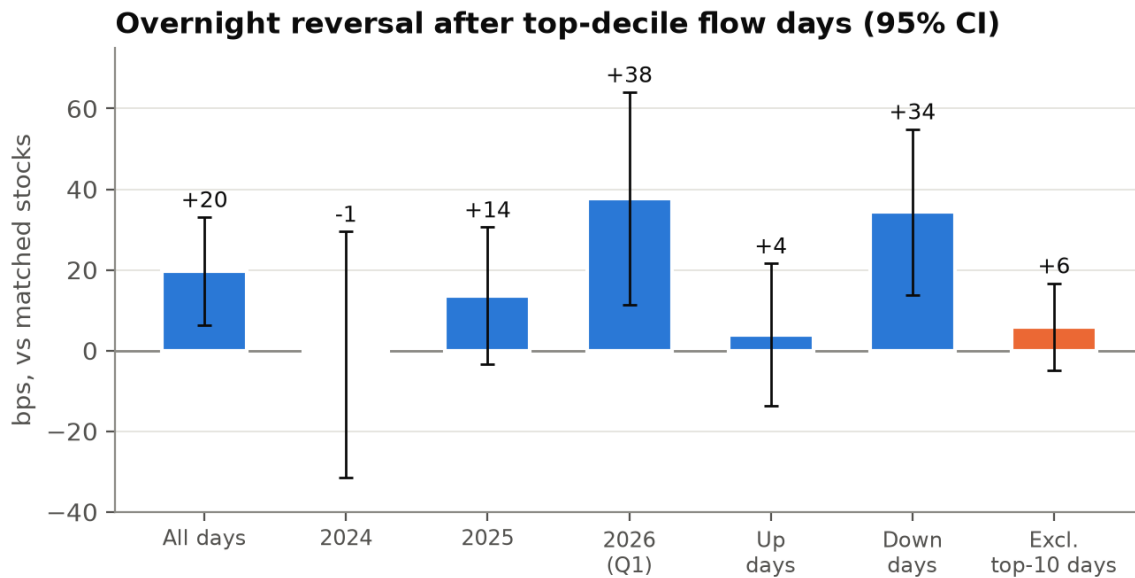


Figure 3. Excess overnight reversal after top-decile flow days vs. same-day matched stocks. It is driven by down days and a handful of dates: the ten biggest days were theme-wide sell-offs in crypto and AI-infrastructure names (COIN, BMNR, CRCL, CLSK, IREN), a factor the QQQ hedge doesn't capture. Without them, the effect is $+6 \pm 11$ bps (95% CI).

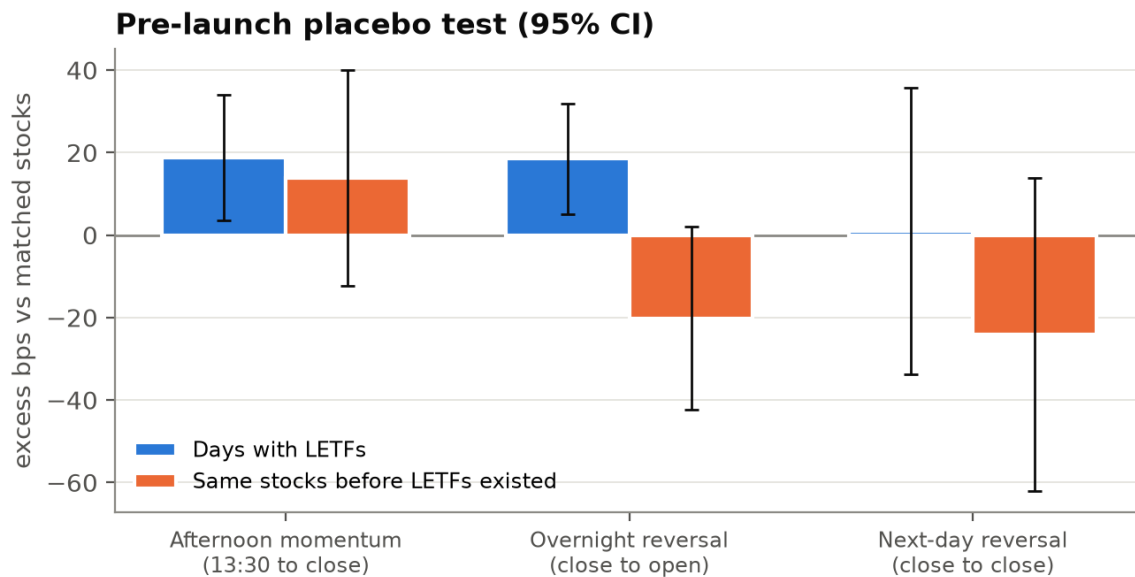


Figure 4. Placebo: the same stocks before any leveraged ETF existed, given the gamma they later had. Afternoon momentum is about as strong without the funds, so it's a trait of volatile retail-favorite stocks, not of the rebalance flow.

What this tells us

- The rebalance flow is public and anticipated. It gets traded into during the afternoon and absorbed before the close, consistent with the published literature and heavy sell-side coverage in 2025-26.
- This matches how the 2007-2011 closing-cross edge faded: once a flow is widely computed, the pressure moves earlier and the close stops being the extreme.
- One correction during the work: an early run showed strong last-30-minute pressure ($t = 6.2$). That came entirely from bad data for ticker B (Barnes Group hourly vs Barrick daily history). After fixing it, the effect is zero.

3. The promising lead: the leveraged ETF's own closing price

Instead of the stock, look at the ETF. At 4:00 its fair value is known almost exactly: $\text{NAV} = \text{yesterday's NAV} \times (1 + \text{leverage} \times \text{the stock's return to its official close})$. If the ETF's own closing price lands away from NAV, and that gap reverts, a liquidity provider can take the other side of the ETF's closing auction and hedge with the stock at the same moment. That is structurally the 2007-2011 trade, in a venue far too small for large firms.

The premium is measured exactly for 38 funds with published daily NAV (GraniteShares, ProShares). For all ~240 funds it uses a proxy, the day's fund return minus leverage $\times$ the stock's return, which correlates 0.69 with the exact measure.

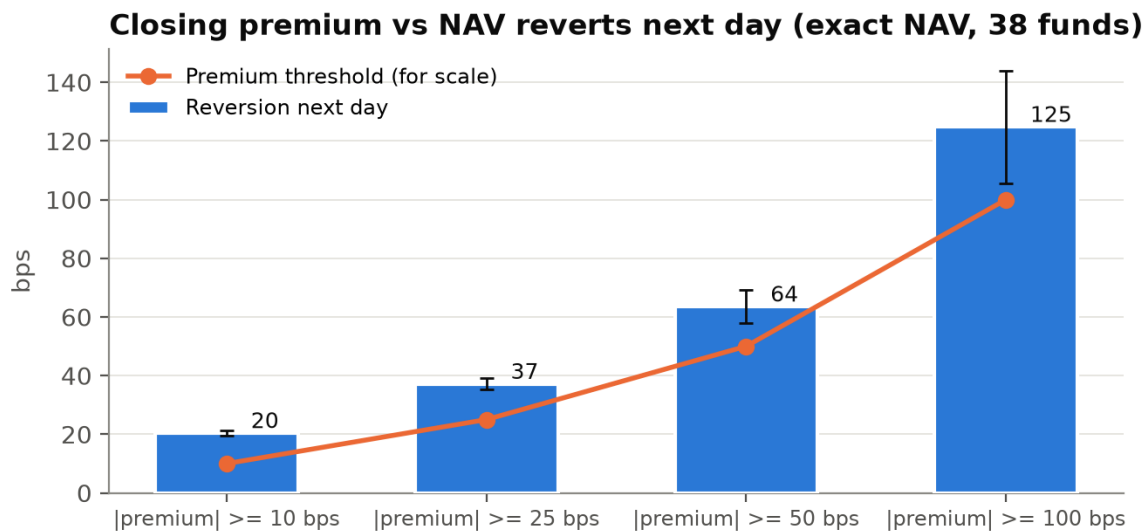


Figure 5. When the ETF closes at a premium or discount to NAV, the gap reverts the next day, almost one-for-one. Exact-NAV funds, July 2022 - March 2026; bars show 95% CI.

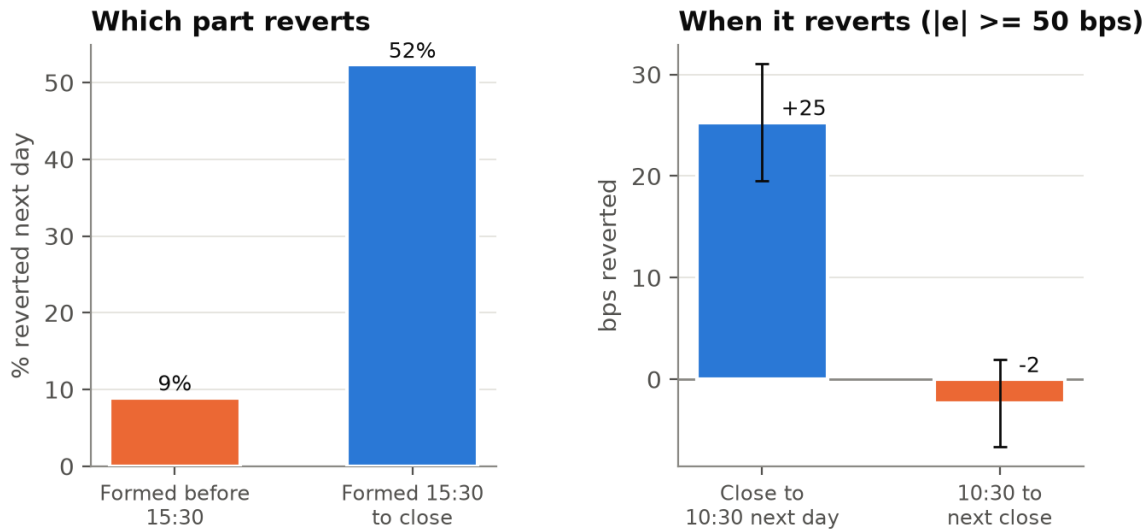


Figure 6. Left: of the day's premium change, the part formed between 15:30 and the close is what reverts (52% next day, vs 9% for the part formed earlier). Right: on large events the reversion is complete by 10:30 the next morning; nothing happens after. Funds with daily trading $\geq \$20M$.

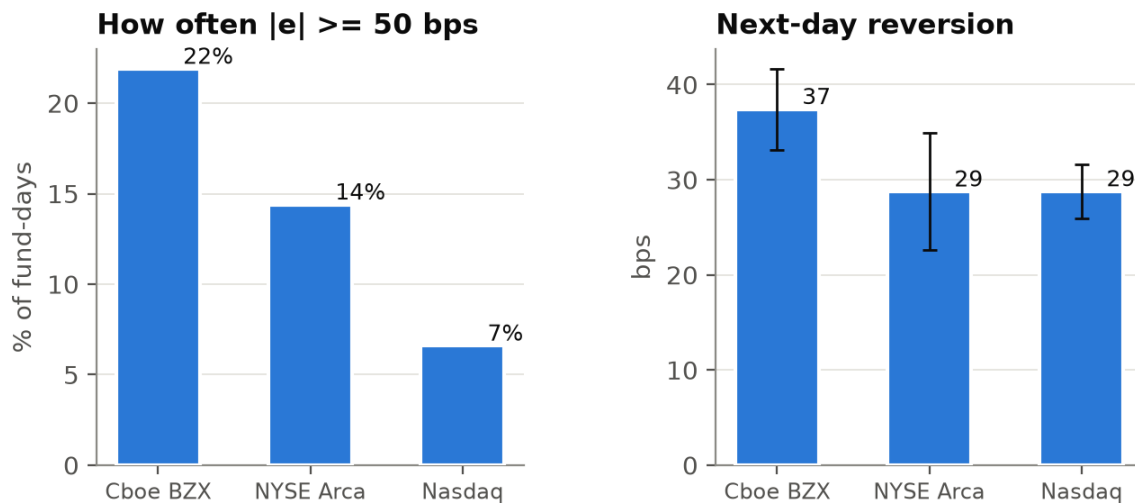


Figure 7. The pattern appears on every listing venue. Cboe BZX listings (mostly T-Rex and Defiance funds) dislocate most often. Not stale prices: the premium barely loads on the stock's move after 15:30 (-0.03 even in the thinnest funds; a stale print would load near -1).

Illustrative P&L

Rule: when the fund's close is at least k bps rich or cheap relative to leverage $\times$ the stock (proxy), take the other side at the close, hedge with the stock at its close, and unwind at the next close. Size = the smaller of \$1M or 1% of the fund's average daily trading. Costs: 5 or 10 bps on the fund leg plus 1 bp on the hedge. **This assumes you can trade at the official closing prices, which is exactly what is not yet verified.**

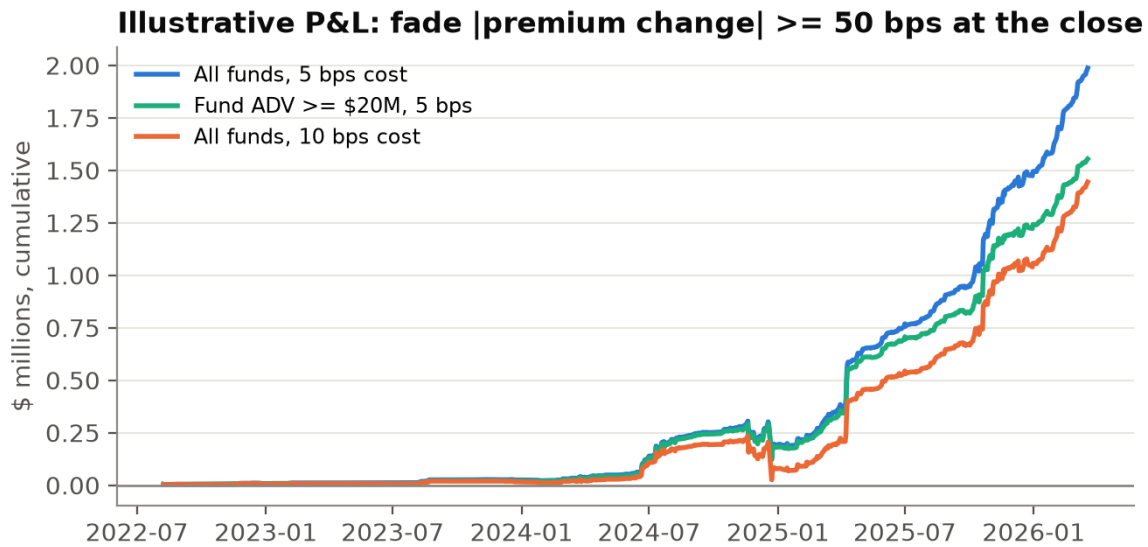


Figure 8. Cumulative P&L of the illustrative rule ($|\text{premium change}| \geq 50$ bps). Most of the P&L comes from 2025 onward, as the number of small single-stock ETFs exploded.

Trigger $ \text{e} $	Cost	Universe	Events	Net / event	P&L total	P&L 2025	Sharpe *	Top-10 days
25 bps	5+1 bps	all funds	15,431	14.1 ± 1.3	\$3.32M	\$1.98M	4.6	22%
50 bps	5+1 bps	all funds	5,627	26.1 ± 2.8	\$1.99M	\$1.30M	3.4	27%
50 bps	10+1 bps	all funds	5,627	21.1 ± 2.8	\$1.44M	\$0.98M	2.5	35%
50 bps	5+1 bps	fund ADV $\geq$ \$20M	1,411	19.2 ± 5.0	\$1.55M	\$1.06M	2.8	34%
50 bps	10+1 bps	fund ADV $\geq$ \$20M	1,411	14.2 ± 5.0	\$1.09M	\$0.79M	2.0	46%
100 bps	5+1 bps	all funds	1,371	46.9 ± 7.1	\$0.81M	\$0.57M	2.2	49%

* Sharpe from daily P&L across all business days, annualized, before hedge basis risk and financing. Net / event shows $\pm$ 95% CI.

4. What could make the lead wrong

- **Executability (the big one).** Yahoo's close is the official closing price. For a thin ETF it can be a small auction print or a last sale at the edge of a wide bid/ask spread. Then the "reversion" is just the price bouncing back to the middle of the spread, and it can't be captured. Everything in section 3 fits a real auction imbalance *and* closing-price noise; only the 4:00 bid/ask and the auction prints tell them apart.
- **Capacity.** ETF closing-auction volume is unknown. 1% of daily trading is a guess and may be optimistic for Cboe-listed funds.
- **Entry timing.** The premium is only known once both auctions print. Live, you'd act on indicative prices from imbalance feeds with limit or imbalance-only orders, so fill rates and adverse selection need modelling.

- **NAV basis.** Funds that hold options (e.g. MSTU in late 2024) or reset swaps off the close make NAV differ from leverage $\times$ stock, adding hedge risk.
- **Multiple testing.** ~50 tests have been run. The lead was motivated by mechanism (handoff item 10), not found by a sweep, but it still needs a single, final test on the untouched hold-out.

Handoff claims that were corrected

Claim	What the sources say
600+ funds, \$41-42B	That's the whole single-stock channel; leveraged + inverse is ~488 funds / \$34B (ETF Action, Sep 2026)
\$50B/day rebalancing record	A peak on some days, mostly index funds; ~\$20B 10-day average (Barclays)
3x single-stock funds	None exist in the US; SEC pushed back on 3x/5x filings (Dec 2025, Mar 2026)
Exposure = leverage $\times$ assets	Often false: MSTR funds were swap-constrained in Nov 2024 and used call options
Nasdaq MOC cutoff 3:50	3:55; Nasdaq imbalance-only orders are repriced to the inside bid/offer
IBKR closing orders	IBKR API has an imbalance-only flag for Nasdaq; no documented NYSE Closing Offset support
Yahoo open = opening cross	Not guaranteed; Yahoo's close is the official close, its open may be the first trade

5. Recommended next steps

Cheapest first

- **1. Bid/ask at 4:00 (free, needs your IBKR session).** Pull 1-minute bid/ask bars for ~30 leveraged ETFs on ~50 large-dislocation days. If the premium still shows at the midpoint, it isn't just bounce.
- **2. Auction prints and imbalances (paid, ask first).** Closing-cross prints and imbalance messages for these ETF tickers (Nasdaq ITCH; Cboe BZX auction feed), e.g. from Databento. They measure auction size and whether our orders would have filled.
- **3. Live recorder (when ready).** Record closing imbalances for the ETFs *and* their stocks 3:50-4:00 daily. This builds the dataset that no historical source provides for free.
- **4. If it holds up:** freeze one simple rule and evaluate it once on the hold-out (23 Mar - 22 Sep 2026). Only after that, consider paper trading.
- **5. Natural extension:** the same NAV-anchored closing-auction trade in other thin ETFs (single-stock option-income, buffer, crypto spot ETFs).

Where to find everything

Code, data caches and the full research log (every variant tried, with results) are in the `kmeans_data` repository, branch `claude/new-session-0kch95`: `RESEARCH_LOG.md`, `reports/*.csv`, `analysis/`, `flows/`, `universe/`.